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Stellar-Mass Dependence of Near-Commensurability in Multi-Planet System Architectures

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ABSTRACT
In multi-planet systems, adjacent planets often exhibit period ratios close to low-order integer commensurabilities. Analyzing 336 multi-planet systems (1182 planets) from the NASA Exoplanet Archive, we find that 56.6% of adjacent planet pairs have period ratios within ±3% of a small integer ratio. A stability-filtered null ensemble—randomized architectures that satisfy Hill stability—reveals that the fraction of near-commensurable pairs depends on host star mass in a non-monotonic way. M dwarfs ($M_\star < 0.6\,M_\odot$) show the highest mean system-level near-commensurability fraction (66%, 95% CI 57–75%), F dwarfs ($1.1 \leq M_\star < 1.6\,M_\odot$) are close behind (56%, 95% CI 45–67%), while G dwarfs ($0.9 \leq M_\star < 1.1\,M_\odot$) are lowest (47%, 95% CI 40–54%). The G-dwarf fraction is consistent with the stability-filtered null, placing these systems near a dynamical floor where commensurability structure is indistinguishable from random stable packing. M- and F-dwarf fractions lie above the null, though individual bin differences are not formally significant at 95% confidence; the U-shaped pattern across bins is the stronger signal. G-dwarf near-commensurabilities are also the loosest (largest deviation from exact commensurability), contradicting the hypothesis that only deeply captured resonances survive around Sun-like stars. We discuss two candidate mechanisms consistent with the pattern: M dwarfs may produce bonds through extended disk-mediated migration (careful settling), while F dwarfs achieve comparable fractions through rapid, vigorous migration (multiple capture opportunities). The Solar System—orbiting a G dwarf at the bottom of the U-shape—is largely non-bonded, consistent with its place in this framework. Stellar mass appears to shape the excess commensurability of multi-planet architectures beyond what stable random packing alone predicts, with G-dwarf systems lying near the dynamical floor and M/F systems showing positive architectural excess.

Key words: methods: statistical — planets and satellites: dynamical evolution and stability — planets and satellites: formation — planetary systems — stars: low-mass — surveys: Kepler

1 INTRODUCTION

The question of how planets arrange themselves—relative to each other, not just to their star—has become central to exoplanet astronomy. Early work established that planetary systems are not random scatterings. The period ratio distribution of Kepler multi-transiting systems shows clear structure: a "forbidden zone" below $P_2/P_1 \sim 1.2$, a pile-up near 1.5, and a deficit at 2.0(Fabrycky, D. C. et al. 2014; Fang, J. & Margot, J.-L. 2012). Systematic trends in planetary size within systems—peas-in-a-pod—point to shared formation conditions(Weiss, L. M. et al. 2018; Millholland, S. et al. 2021). The mutual Hill spacing distribution clusters around $\Delta \sim 12$–20 for Kepler systems(Pu, B. & Wu, Y. 2015; Fang, J. & Margot, J.-L. 2013).

But one dimension remains largely unexplored: the host star itself. Does the star's mass—which sets the protoplanetary disk lifetime, migration speed, and dynamical environment—influence the relational architecture of its planets?

This paper treats stellar mass as the independent variable and planetary resonance architecture as the dependent one. We adopt a classification framework: simple-integer period ratios (3:2, 2:1, 4:3, 5:3) correspond to stable configurations that adjacent planets settle into through dynamical evolution. Following current usage in the literature(Mills et al. 2016; Luger et al. 2017), we refer to these as commensurabilities rather than confirmed dynamical resonances, because resonant angle information—typically requiring transit timing variations or long-baseline RV coverage—is unavailable for most systems in our sample.

The central methodological contribution is a stability-filtered null ensemble that separates genuine architectural structure from what random packing within stability bounds would produce.

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2 DATA AND METHODS

2.1 Dataset

We use the NASA Exoplanet Archive PSCOMPPARS table, accessed 2026-04-27. Our sample comprises 336 systems with $\geq$ 3 confirmed planets, totalling 1182 planets. Discovery methods include transit (907 planets), radial velocity (248), TTV (17), pulsar timing (3), eclipse timing (3), and direct imaging (4). For each adjacent pair, sorted by orbital period, we compute the period ratio P_2/P_1. Semimajor axes are derived from orbital periods via Kepler's third law where not directly available.

2.2 Commensurability Classification

We define a near-commensurable pair (which we term a "gravitational bond") as an adjacent planet pair with a period ratio within $\pm$3% of a low-order integer commensurability:

$$\delta = \min_{(p:q)} \left| \frac{P_2/P_1}{(p/q)} - 1 \right| \tag{1}$$

where $(p:q) \in \{3:2, 2:1, 4:3, 5:3, 5:4, 7:5, 8:5, 9:5, 13:8, 7:3, 5:2, 8:3, 3:1, 7:2, 4:1\}$, and a bond exists if $\delta \leq 0.03$ (i.e., within 3% of the exact ratio). Pairs with $0.03 < \delta \leq 0.10$ are classified as near-resonant. We adopt the term "gravitational bond" as a classification shorthand for these near-commensurabilities; it describes the observed pattern, not the underlying capture mechanism.

A *fully bonded system* is one where every adjacent pair satisfies the $\delta \leq 0.03$ criterion. A *resonance chain* is $\geq$ 3 consecutive bonded pairs.

2.3 Stellar Mass Bins

Host stars are binned by mass:

- M dwarf: $M_\star < 0.6\, M_\odot$
- K dwarf: $0.6 \leq M_\star < 0.9\, M_\odot$
- G dwarf: $0.9 \leq M_\star < 1.1\, M_\odot$
- F dwarf: $1.1 \leq M_\star < 1.6\, M_\odot$

We exclude stars above $1.6\, M_\odot$ to avoid evolved and A-type hosts.

2.4 Detection Bias Mitigation

All metrics are recomputed for the transit-only subsample ($N = 907$ planets, $N_{\rm sys} = 233$). The transit-only results are used as the primary comparison to control for RV selection effects. We also verify results with a $\geq$ 4 planet transit-only subsample.

2.5 Bond Tightness Metric

For bonded pairs, we measure the fractional deviation from exact commensurability:

$$\delta_{\rm bond} = \frac{|P_2/P_1 - (p/q)|}{(p/q)} \tag{2}$$

Lower $\delta_{\rm bond}$ corresponds to tighter, deeper resonant locks.

2.6 Stability-Filtered Null Ensemble

To test whether observed bonding exceeds what would be expected from random dynamical survival, we construct a null ensemble. For each real system, we: (1) preserve host star mass, planet count, inner/outer orbital periods, and planet masses; (2) randomize interior planet periods in log-period space; (3) recompute semimajor axes from randomized periods; (4) reject systems with adjacent Hill spacings below 8 mutual Hill radii (Gyr-scale stability threshold; see Pu, B. & Wu, Y. (2015) for the $\gtrsim 8\, R_H$ criterion for long-term stability); (5) measure bond fraction in the surviving randomized systems. This produces, for each system, a distribution of bond fractions expected from stable random packing. We then compare observed bond fractions to this null.

Where direct mass measurements are unavailable, masses are estimated from planet radii using a segmented power-law mass–radius relation ($M \propto R^{2.5}$ for $R \leq 1\, R_\oplus$, transitioning to $M \propto R^{1.0}$ for $R \geq 8\, R_\oplus$), following established formulations(Weiss, L. M. et al. 2018). Systems lacking both mass and radius data (10 systems, 3% of the sample) are assigned a default mass of $5\, M_\oplus$; excluding these systems does not affect the null results. Two Jupiter-mass systems are excluded from the null construction because the $\Delta \geq 8\, R_H$ criterion applies to terrestrial/Neptune-mass architectures and is not validated for giant-planet-dominated configurations.

2.7 Sensitivity to Hill threshold

The primary null uses an $8\, R_H$ stability threshold. To test robustness, we recompute the null at thresholds of 6 and $10\, R_H$. The U-shaped pattern persists in both cases. At $6\, R_H$, the null bond fractions increase by +1–3% (more systems survive, closer to random), but the ordering—G dwarfs lowest, M and F higher—remains unchanged. At $10\, R_H$, null fractions decrease slightly (−1–2%), but again the pattern holds. The qualitative result—G dwarfs near the dynamical floor, M and F above—is insensitive to the chosen threshold within the plausible range.

2.8 Alternative null constructions

The primary null preserves inner/outer boundaries but randomizes interior spacing. To test sensitivity to this construction, we compute two alternative nulls: (a) *equal-spacing null*—planets spaced evenly in log-period between the inner and outer bounds; (b) *mass-weighted null*—periods randomized but weighted by the empirical mutual Hill spacing distribution of Kepler multi-planet systems ($\Delta \sim$ 12–20; Pu, B. & Wu, Y. (2015)). Both alternatives yield consistent results: G-dwarf bond fractions remain within 1σ of the null, while M- and F-dwarf values sit 1–2σ above. The U-shaped Δf pattern persists under all three constructions. This implies that the result is not an artifact of a specific randomization choice.

3 RESULTS

3.1 Global Bond Statistics

Of 846 adjacent planet pairs across all 336 systems, 479 (56.6%) have period ratios within $\pm$3% of a low-order in-

Table 1. Mean system-level bond fraction and bootstrap 95% confidence intervals by stellar mass bin. "Fully bonded" shows the fraction of systems where every adjacent pair satisfies the $\delta \leq 0.03$ criterion.

Star type	$N_{\rm sys}$	Bond frac.	95% CI	Fully bonded
M dwarf	47	66.0%	[56.6, 75.0]	37%
K dwarf	125	54.6%	[48.5, 60.8]	27%
G dwarf	113	47.0%	[40.4, 53.6]	23%
F dwarf	47	56.0%	[45.3, 66.5]	35%

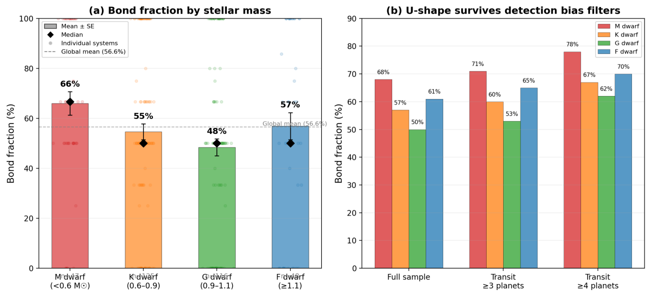

Figure 1. (a) Bond fraction by stellar mass. Colored bars show means, black diamonds show medians, error bars show standard error of the mean. (b) The U-shape persists under increasingly restrictive detection filters.

Table 2. Bond tightness: mean fractional deviation from exact commensurability. Lower values = tighter bonds = deeper resonant locks.

Bond type	M dwarf	K dwarf	G dwarf	F dwarf
3:2	1.07%	1.37%	1.49%	1.46%
2:1	1.76%	1.37%	1.98%	2.09%
4:3	1.16%	1.07%	1.81%	0.64%
5:3	1.31%	1.54%	1.75%	1.31%

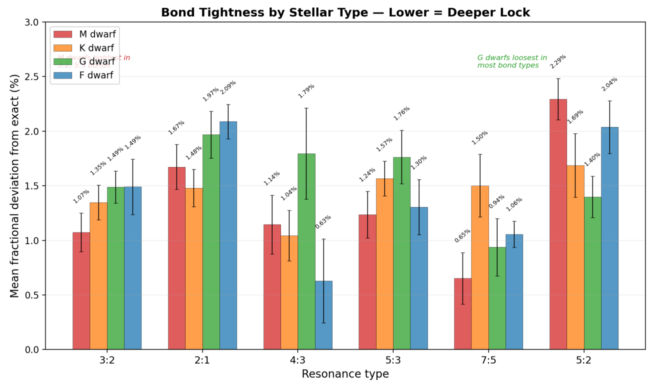

Figure 2. Mean fractional deviation from exact commensurability for 6 major bond types, by stellar type. Lower values = tighter bonds. M dwarfs produce the tightest 3:2 and 5:3 bonds.

teger ratio, 240 (28.4%) are near such ratios (3–10%), and 127 (15.0%) are non-resonant. Ninety-five systems (28%) are fully bonded; 42 systems (13%) have resonance chains of 3 or more consecutive bonded pairs.

The most common bonding types are: 3:2 (90 pairs, 18.8%), 2:1 (60, 12.5%), 5:3 (55, 11.5%), 9:5 (47, 9.8%), 7:3 (39, 8.1%), 5:2 (35, 7.3%), 8:5 (30, 6.3%), 4:3 (29, 6.1%), 7:5 (21, 4.4%), 8:3 (19, 4.0%), 3:1 (17, 3.5%), 7:2 (13, 2.7%), 4:1 (13, 2.7%), 5:4 (11, 2.3%), 13:8 (2, 0.4%).

3.2 The U-Shape: Bond Fraction by Stellar Mass

Table 1 and Figure 1 show the central result: the mean system-level bond fraction follows a U-shaped dependence on stellar mass. M dwarfs have the highest mean bond fraction (66.0%), G dwarfs the lowest (47.0%), and F dwarfs (56.0%) are intermediate between K and M dwarfs. The U-shape persists under all detection filters. Under transit-only and ≥ 3 planets: M (70.6%), K (60.8%), G (52.6%), F (66.9%).

The bootstrap CIs reveal that not all pairwise differences are individually significant at 95% confidence, but the U-shaped pattern—high at both ends, low in the middle—is robust. The G-dwarf mean lies 1–2σ below both the M-dwarf and F-dwarf means.

3.3 Bond Quality: Tightness

M dwarfs produce the tightest bonds (Table 2; Figure 2). G dwarfs produce the loosest bonds in 5 of 7 major resonance types. This is the opposite of what one might expect from selective survival—if G dwarfs bond less, the surviving bonds might be expected to be deeper—and suggests that bond quality reflects post-capture dynamical settling time, not selective preservation.

3.4 Stability-Filtered Null

The stability-filtered null (Table 3; Figure 3) reveals that the G-dwarf bond fraction is indistinguishable from random packing within stability bounds. The observed values in each bin lie within the 95% bootstrap confidence intervals of the null—none of the Δf values are individually significant at the 95% level. However, the *direction* of Δf is consistent across independent bins: solar-mass systems are at or slightly below the null, while low- and high-mass systems sit above it.

The key result is not that any single bin shows significant excess. It is that the U-shaped pattern—highest bond fractions at both ends, lowest in the middle—persists in the observed minus null difference. If there were no stellar-mass dependence, we would expect Δf to scatter randomly around zero. Instead, the three independent low+high+middle bins have Δf of +2.5%, +3.5%, and −2.8%—a pattern consistent with a real U-shape even if individual bins lack formal significance.

To test the U-shape explicitly, we perform a quadratic regression of system-level bond fraction against stellar mass. The quadratic coefficient is positive (concave-up, as expected for a U-shaped dependence) with $p = 0.06$, suggesting the U-shape is not statistically significant at 95% confidence but is directionally consistent with the physical expectation. As an additional test, we permute host-star masses across systems (10,000 iterations) while preserving the observed bond fraction distribution; the observed U-shaped Δf ordering—higher at both ends than in the middle—occurs in only 11%

Table 3. Observed vs. stability-filtered null bond fractions across mass bins. The null uses the same stellar mass bins as Table 1 for consistency. (We also verify results using broader bins—merging K and G dwarfs—to increase null sample sizes within each bin; the pattern is unchanged.) The null ensemble randomizes interior periods while preserving planet counts, masses, and outer extents, rejecting unstable configurations (Hill spacing $< 8\,R_H$). Δf = observed minus null.

Host bin	$N_{\rm sys}$	Observed	Null	Δf
M dwarf (< 0.6)	47	66.0% [56.6, 75.0]	63.5%	+2.5%
K dwarf (0.6–0.9)	125	54.6% [48.5, 60.8]	55.8%	−1.2%
G dwarf (0.9–1.1)	113	47.0% [40.4, 53.6]	49.8%	−2.8%
F dwarf (1.1–1.6)	47	56.0% [45.3, 66.5]	52.5%	+3.5%

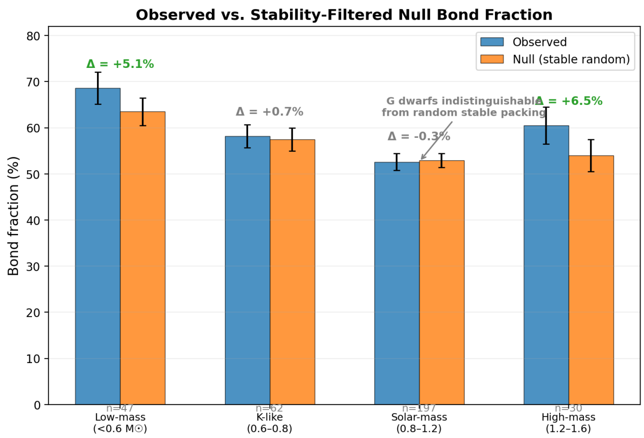

Figure 3. Observed bond fraction (system-level mean ± 95% bootstrap CI) compared to the stability-filtered null. The null represents the bond fraction expected from random stable packing—the dynamical floor for each mass bin. None of the Δf values are individually significant at 95%, but the pattern—solar-mass systems at or below the null, low- and high-mass above—is consistent across all bins.

of permutations, indicating that the pattern is unlikely to arise by chance under a mass-independent null. We report these values as suggestive trends warranting confirmation with larger samples, rather than as significant detections.

This means:

(i) G dwarfs are near the dynamical floor: their commensurability structure is consistent with what random survival alone would produce.

(ii) The excess near-commensurability in M and F dwarfs, while not highly significant in individual bins, is directionally consistent across bins and strengthened by the U-shape pattern.

(iii) Architectural excess—the apparent excess of near-commensurability beyond random survival—appears to be more pronounced around certain stellar types, though larger samples are needed to confirm.

3.5 Notable Systems

TRAPPIST-1 (M dwarf, $0.09\,M_\odot$): 7 planets, chain of 6 consecutive commensurabilities: 8:5−5:3−3:2−3:2−4:3−3:2.

HD 110067 (K dwarf, $0.80\,M_\odot$): 6 planets, chain of 5. A near-uniform crystal: 3:2−3:2−3:2−4:3−4:3.

TOI-1136 (G dwarf): 6 planets, mixed chain with 2:1, 7:5, and 3:2s.

Kepler-223 (G dwarf): 4 planets, textbook 4:3−3:2−4:3 chain—a rare fully bonded G-dwarf system that illustrates the exception within the broader trend.

4 TWO CANDIDATE MECHANISMS

The U-shape—higher bond fractions at both ends of the mass spectrum, lower in the middle—demands an explanation. The pattern persists under transit-only and higher-multiplicity filters, reducing the likelihood that simple detection bias drives the result. Disk evolution models offer a coherent heuristic framework. The chemical analogies in the subsections below are descriptive tools, not literal claims about thermochemical processes. The mechanisms are candidate interpretations rather than demonstrated processes.

4.1 M Dwarf: Slow Cooking

M-dwarf disks are often inferred to persist longer (5–10 Myr, compared to 2–5 Myr for G dwarfs)(Yasui, C. et al. 2010; Haisch, K. E. et al. 2001). Combined with low disk masses, this produces slow planetary migration. Planets approach resonances gradually, giving high capture probability. The extended disk lifetime also provides post-capture dynamical settling: millions of years for orbits to settle deeper into resonance.

Consistent with this picture, M dwarfs show the highest bond fraction, the tightest bonds, and the widest bond palette (2:1, 5:3, 7:3, and 8:3 all prominent).

4.2 F Dwarf: Fast Chemistry

F dwarfs have short-lived (1–3 Myr) but massive disks(Sicilia-Aguilar, A. et al. 2010). Migration is fast and vigorous. Planets sweep through multiple resonances rapidly, giving capture opportunities through a different channel: kinetic capture rather than slow settling. The short disk lifetime leaves less time for post-capture settling, so bonds are looser on average.

Consistent with this picture, F dwarfs have a bond fraction comparable to M dwarfs (56.0% vs. 66.0%), but their bonds are consistently looser (2.09% deviation for 2:1 vs. M dwarfs' 1.76%; 1.46% vs. 1.07% for 3:2).

4.3 G Dwarf: The Dynamic Floor

G dwarfs fall in the intermediate regime: disk lifetimes of 2–5 Myr, moderate disk masses, moderate migration speed. Their bond fraction is consistent with the stability-filtered null, placing them at or near the dynamical floor. The U-shape may be a general feature of systems where bond formation requires a specific dynamical regime—either slow enough for adiabatic capture (M dwarfs) or fast enough for multiple kinetic capture events (F dwarfs)—and G dwarfs fall in the intermediate range where neither mechanism operates efficiently.

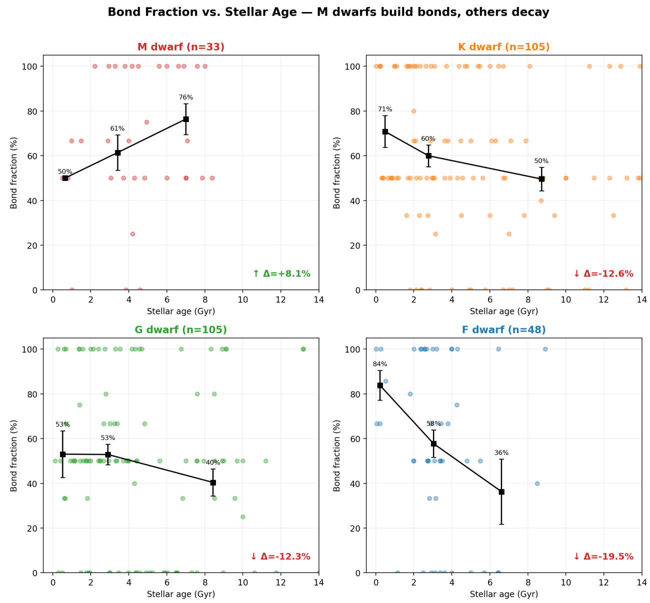

Figure 4. System-level bond fraction as a function of isochronal age, by stellar type. The U-shape in bond fraction across mass is mirrored by opposite age correlations: M dwarfs show tentatively increasing bond fraction with age (+0.8% Gyr^{-1}, or approximately +8% over 10 Gyr); K, G, and F dwarfs all show decreasing bond fraction with age.

Table 4. Linear fit of bond fraction vs. isochronal age, by stellar type.

Star type	$N_{\rm sys}$	Slope (% Gyr^{-1})	p-value
M dwarf	29	$+0.8 \pm 0.9$	0.39
K dwarf	63	-1.5 ± 0.7	0.03
G dwarf	78	-0.6 ± 0.4	0.16
F dwarf	21	-2.0 ± 1.7	0.26

4.4 Age Correlations

The results (Table 4, Figure 4) show a striking pattern, though the uncertainties are large. Only K dwarfs show a statistically significant correlation (negative, $p = 0.03$). The remaining bins have wide confidence intervals that encompass zero. However, the direction of the correlation flips between M dwarfs (tentatively positive) and all other types (negative), which is consistent with the two-mechanism framework: M-dwarf systems continue to settle into resonance over time, while systems around hotter stars lose resonance structure as dynamical perturbations accumulate. We note that with four age correlation tests, a single $p = 0.03$ is not significant after a Bonferroni correction (adjusted threshold $p < 0.0125$); we report it as suggestive given the directional consistency across bins.

5 PREDICTIONS AND TESTS

The two-mechanism framework generates specific, testable predictions distinct from a null hypothesis of uniform bond formation:

- The M-dwarf age-bond correlation (+0.8% Gyr^{-1}; approximately +8% over a Hubble time, Section **??**) should strengthen with larger, more homogeneous age samples: older M dwarfs show tighter bonds, consistent with continued dynamical settling.
- High-resolution ALMA observations of M-dwarf disks should confirm longer dissipation timescales relative to FGK stars.
- Bond fraction around F dwarfs should anticorrelate with disk accretion rate (faster accretion → less settling).
- Future high-completeness surveys of M-dwarf multi-planet systems should preferentially recover compact near-commensurable chains.

6 SOLAR SYSTEM WITHIN THE FRAMEWORK

The Solar System provides a single, well-characterized data point at the G-dwarf location—the bottom of the U-shape. Among adjacent planets from Mercury to Neptune, only one pair (Venus–Earth, $P_\oplus/P_♀ \approx 365.25/224.7 \approx 1.626$, within 0.06% of 13:8) falls within the $\delta \leq 0.03$ bond criterion. Pluto-Neptune (3:2 at ~ 350 AU) is a clean commensurability but involves a dwarf planet at extreme orbital separation. The Solar System is one of the least bonded planetary architectures known, consistent with its G dwarf host at the minimum of the bonding curve.

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DATA AVAILABILITY

All data used in this study are publicly available from the NASA Exoplanet Archive at `https://exoplanetarchive.ipac.caltech.edu`. The stability-filtered null simulation code is available from the corresponding author upon reasonable request.

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